

Virtual Networks & Subnets: Building Your Private Network

The foundational building blocks every Azure resource needs to live and communicate

● Beginner

🕒 15 Minutes

📘 AZ-104 · Module 04

Why This Matters

Before you can deploy any Azure resource (VM, database, app service), it needs a network to live in. Virtual Networks are like creating your own private network inside Azure — like a gated community where only authorized traffic can enter. Understanding VNets and subnets is foundational to everything else in Azure networking.

Before You Start

Prerequisites	00-Prerequisites: Networking Basics (CIDR, IP addresses, subnets)
Time to understand	15 minutes
Difficulty	● Beginner (foundational, no prior network knowledge needed)
What you'll learn	What VNets are, how to create subnets, why they matter

The Simple Idea

What Is a Virtual Network (VNet)?

A **VNet** is your own **private network inside Azure**. It's a walled-off space where your resources live, talk to each other, and stay isolated from the internet (unless you explicitly allow internet access).

What Is a Subnet?

A **subnet** is a **smaller network inside a VNet**. You divide your VNet into subnets to organize resources and apply security rules.

Each subnet has different security rules (Network Security Groups).

Without a VNet (insecure):

Your computers sitting on public internet

↓

Anyone can see them and try to connect

↓

No privacy, high risk

With a VNet (secure):

Your office building (VNet)

├ Reception area (public IP - can access internet)

├ Employee cubicles (private IPs - internal only)

├ Server room (no public IP - completely isolated)

└ Conference room (VPN access only - external partners)

Locked doors (Network Security Groups) control who goes where

Example VNet structure:

VNet: 10.0.0.0/16 (my company network)

├ Subnet 1: 10.0.1.0/24 (Web tier - public)

├ Subnet 2: 10.0.2.0/24 (App tier - private)

└ Subnet 3: 10.0.3.0/24 (Database tier - completely private)

How It Actually Works

Step 1: Create a VNet

You define an address space (like your company's IP range):

VNet Name: my-company-network

Address Space: 10.0.0.0/16

(This gives you 65,536 possible IP addresses)

Step 2: Divide Into Subnets

You create smaller subnets within the VNet:

```
Subnet 1 (Web): 10.0.1.0/24 (256 addresses)
Subnet 2 (App): 10.0.2.0/24 (256 addresses)
Subnet 3 (DB): 10.0.3.0/24 (256 addresses)
```

Step 3: Deploy Resources Into Subnets

```
Subnet 1 (Web): VM with public IP
                 (accessible from internet)
Subnet 2 (App): VM with private IP
                 (internal only)
Subnet 3 (DB):  SQL Database
                 (no public IP)
```

Step 4: Apply Security Rules (NSGs)

```
Subnet 1: "Allow port 80 from internet"
Subnet 2: "Allow port 5000 from Subnet 1 only"
Subnet 3: "Allow port 1433 from Subnet 2 only"
```

Result: Traffic flows securely through your network tiers.

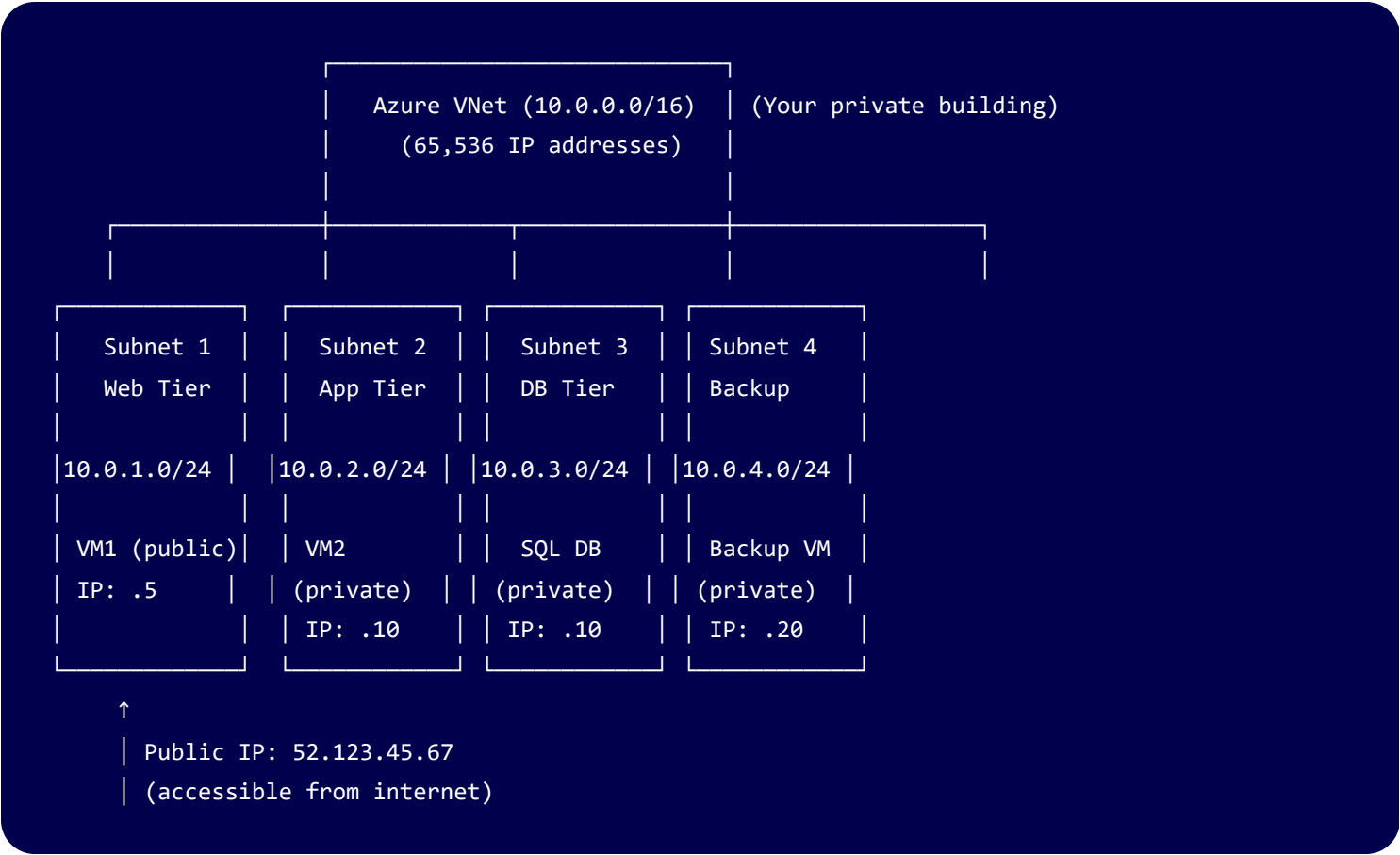
Key Components: Understanding VNet Structure

Component	What It Is	Example	Why It Matters
VNet	Your private network	10.0.0.0/16	Creates isolated space for resources
Address Space	Range of IP addresses	10.0.0.0/16 (65k IPs)	Defines what IPs you can assign

Subnet	Smaller network within VNet	10.0.1.0/24 (256 IPs)	Organize resources by tier/function
Public IP	Internet-accessible address	52.123.45.67	Allows VM to receive traffic from internet
Private IP	Internal-only address	10.0.2.5	Resources talk to each other internally
Network Security Group (NSG)	Firewall rules	"Allow port 80 inbound"	Controls traffic in/out of subnet

Mental Model: VNet as a Building

Think of your VNet like a **building with security**:



Each subnet has security rules (like locked doors):

- **Subnet 1 (Web):** Door open to internet (port 80)

- **Subnet 2 (App):** Door locked except from Subnet 1
- **Subnet 3 (DB):** Door locked except from Subnet 2
- **Subnet 4 (Backup):** Door locked — internal only

Worked Example: Real Scenario

The Scenario

TechCorp is building a 3-tier web application:

- **Web Tier:** User-facing web servers (needs internet access)
- **App Tier:** API servers (talks to web tier and database)
- **Database Tier:** SQL databases (talks to app tier only)

Goal: Deploy this securely with separate tiers, each with appropriate access.

Step 1: Plan the VNet

VNet Address: 10.0.0.0/16

Web Subnet: 10.0.1.0/24 (256 IPs for web servers)
App Subnet: 10.0.2.0/24 (256 IPs for app servers)
DB Subnet: 10.0.3.0/24 (256 IPs for databases)

Step 2: Create Subnets & Deploy Resources

Web Subnet (10.0.1.0/24)

VM: WebServer1
├ Private IP: 10.0.1.5
├ Public IP: 52.123.45.67
| ← Internet can reach here
└ Purpose: Accept web traffic from internet

App Subnet (10.0.2.0/24)

```
VM: AppServer1
├ Private IP: 10.0.2.10
├ No public IP
├   ← Internet CANNOT reach here
└ Purpose: Talk to web tier and
  database only
```

DB Subnet (10.0.3.0/24)

```
Database: SQL Server
├ Private IP: 10.0.3.20
├ No public IP
├   ← Internet CANNOT reach here
└ Purpose: Accept queries from
  app tier only
```

Step 3: Apply Security Rules (Network Security Groups)

Web Subnet Security Rules (NSG)

```
✓ ALLOW: Port 80 (HTTP) from internet
✓ ALLOW: Port 443 (HTTPS) from internet
X DENY: Everything else from internet
✓ ALLOW: App tier to reach web tier
```

App Subnet Security Rules (NSG)

```
✓ ALLOW: Port 5000 from web tier
✓ ALLOW: Database tier to reach app tier
X DENY: Internet access (no public IP)
```

DB Subnet Security Rules (NSG)

```
✓ ALLOW: Port 1433 (SQL) from app tier
X DENY: Everything else
```

X DENY: Internet access (no public IP)

Step 4: Traffic Flow

User requests web page:

1. User on internet → WebServer1 (52.123.45.67)
✓ Allowed (public IP, port 80 open)
2. WebServer1 → AppServer1 (10.0.2.10)
✓ Allowed (internal subnet communication)
3. AppServer1 → SQL Server (10.0.3.20)
✓ Allowed (port 1433 from app tier)
4. SQL Server → Internet
X Blocked (no public IP, NSG denies)

Security achieved: Database is unreachable from internet, even if someone compromises the web server.

Important Concept: Azure Reserved IPs

Important

Azure reserves 5 IPs in every subnet.

Formula:

Usable IPs = Total IPs - 5
Total IPs = $2^{(32 - \text{subnet_mask})}$

Example: /24 subnet
Total = $2^{(32-24)} = 2^8 = 256$
Usable = $256 - 5 = 251$

IP	Purpose
.0	Network address (reserved)
.1	Default gateway (reserved)
.2	Azure DNS (reserved)
.3	Azure future use (reserved)
.255	Broadcast address (reserved)

Example for subnet 10.0.1.0/24 (256 IPs):

```
Total IPs: 256
Reserved: 5 (listed above)
Usable: 256 - 5 = 251 IPs for your resources
```

Common Mistakes (What NOT to Do)

✗ Mistake 1: Overlapping Address Spaces

Wrong

```
VNet 1: 10.0.0.0/16
VNet 2: 10.0.0.0/16 (same range!)

If you peer these VNets:
↓
Network conflict-traffic routing fails
↓
Applications can't communicate
```

Fix

```
Plan address spaces before creating VNets:
VNet 1: 10.0.0.0/16
```



```
VNet 2: 10.1.0.0/16 (different range)
```

```
VNet 3: 10.2.0.0/16 (different range)
```

Document them:

```
|— Finance VNet: 10.0.0.0/16
```

```
|— Engineering VNet: 10.1.0.0/16
```

```
|— Operations VNet: 10.2.0.0/16
```

Why it fails: Azure can't route traffic to same IP range in two places.

✗ Mistake 2: Subnet Too Large

Wrong

```
VNet: 10.0.0.0/16
```

```
Single subnet: 10.0.0.0/16 (all 65k IPs in one subnet)
```

Result:

```
|— No tier separation
```

```
|— All resources see all traffic
```

```
|— Hard to apply targeted security
```

```
|— Scaling becomes difficult
```

Fix

Divide into multiple smaller subnets:

```
VNet: 10.0.0.0/16
```

```
|— Web: 10.0.1.0/24 (256 IPs)
```

```
|— App: 10.0.2.0/24 (256 IPs)
```

```
|— DB: 10.0.3.0/24 (256 IPs)
```

```
|— Extra: 10.0.4.0-10.0.254.0/24 (for growth)
```

Why it fails: Can't separate concerns (web vs. app vs. database).

✗ Mistake 3: Forget About Reserved IPs

Wrong

```
Subnet: 10.0.1.0/24
```

```
Admin thinks: "I have 256 IPs, let me assign 256 VMs"
```

```
↓
```

```
Creates VMs with IPs: 10.0.1.0, 10.0.1.1, ..., 10.0.1.255
```

↓
Fails: IPs .0, .1, .2, .3, .255 are reserved
↓
Only 251 VMs can be created, not 256

Fix

Calculate usable IPs:
/24 subnet = 256 total - 5 reserved = 251 usable
/25 subnet = 128 total - 5 reserved = 123 usable
/26 subnet = 64 total - 5 reserved = 59 usable

Plan accordingly for expected scale.

Why it fails: Reserved IPs aren't available for resources.

How This Connects to Other Topics

Related to Module 01 (Identity & Governance)

- **RBAC + VNets:** Use RBAC to control who can create/modify VNets
- **Policy + VNets:** Enforce policies like "all VNets must have NSGs"

Related to Module 02 (Storage)

- **Service Endpoints:** Connect storage accounts to VNet without public IP
- **Private Endpoints:** Access storage securely from VNet
- **Firewall rules:** Restrict storage access to specific VNets

Related to Module 03 (Compute)

- **VM Deployment:** VMs must be in a subnet (the network they live in)
- **Load Balancers:** Distribute traffic to VMs across subnets

Related to Module 05 (Monitor)

- **Network Watcher:** Monitor traffic between subnets
- **Flow Logs:** See which resources are communicating with each other

See It In Action

Associated lab: Lab 12: Create & Configure VNets

Suggested learning sequence:

1. ☒ Read this doc (VNets & Subnets - foundation)
2. ☒ Read **Network Security Groups** (how to secure them)
3. ☒ Read Routing Fundamentals (how traffic flows)
4. ☒ Work through Lab 12 (hands-on VNet creation)
5. ☒ Read VNet Peering (connect multiple VNets)

Key Takeaways



Summary

- **VNet = Your private network in Azure** (like an office network)
- **Subnet = Division within VNet** (like separate departments)
- **Each resource needs a VNet and subnet** (fundamental requirement)
- **Plan address spaces ahead** (avoid overlaps and conflicts)
- **Reserve 5 IPs per subnet** (don't count them in usable IPs)
- **Multiple subnets = better security** (separate tiers, different rules)
- **Public IP = Internet accessible** (VMs with public IPs can reach internet)
- **Private IP = Internal only** (VMs without public IPs stay private)

Summary: VNet vs. Subnet vs. Resource Group

Aspect	VNet	Subnet	Resource Group
Purpose	Create private network	Organize within VNet	Organize any resources
Contains	Subnets, resources	Resources (VMs, etc.)	Any Azure resources
Region	Region-specific	Within VNet (same region)	Can span regions
Required	Yes (for networking)	Yes (to deploy resources)	Yes (to organize resources)
IP space	Address space you define	Divided from VNet	N/A (no IP concept)

Next Steps

1. **Learn:** Read this doc (you're here)

- 2. **Secure:** Read **Network Security Groups** (how to add firewall rules)
- 3. **Understand:** Read Routing Fundamentals (how traffic flows)
- 4. **Practice:** Lab 12: Create & Configure VNets
- 5. **Connect:** Read VNet Peering (connect multiple VNets)

Azure AZ-104	Module	Generated
Professional Concept Documentation	04 — Networking	June 15, 2026

© 2026 — AZ-104 Learning Hub. For study purposes only.